

Qatar-2b

R-0929

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_180513 · created 2026-07-29T11:01:07+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458251.7001 +/- 0.00058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1973 +/- 0.0097
Transit depth (Rp/Rs)^2	3.89 +/- 0.38 [%]
Orbital Inclination (inc)	86.15 +/- 1.09
Airmass coefficient 1 (a1)	1.016 +/- 0.015
Airmass coefficient 2 (a2)	-0.012 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.48 %
Best Comparison Star	#1 - [138, 211]
Optimal Aperture	2.42
Optimal Annulus	10.36
Transit Duration (day)	0.0737 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1973 $\pm$ 0.0097 vs 0.16327 (deviation 2.72 σ)
Duration fit vs published	0.0737 d vs 0.0767 d (deviation 0.68 σ)
Mid-time O-C	-1.5 min
Depth SNR	15.8
Residual scatter	1.48 %

Light curve

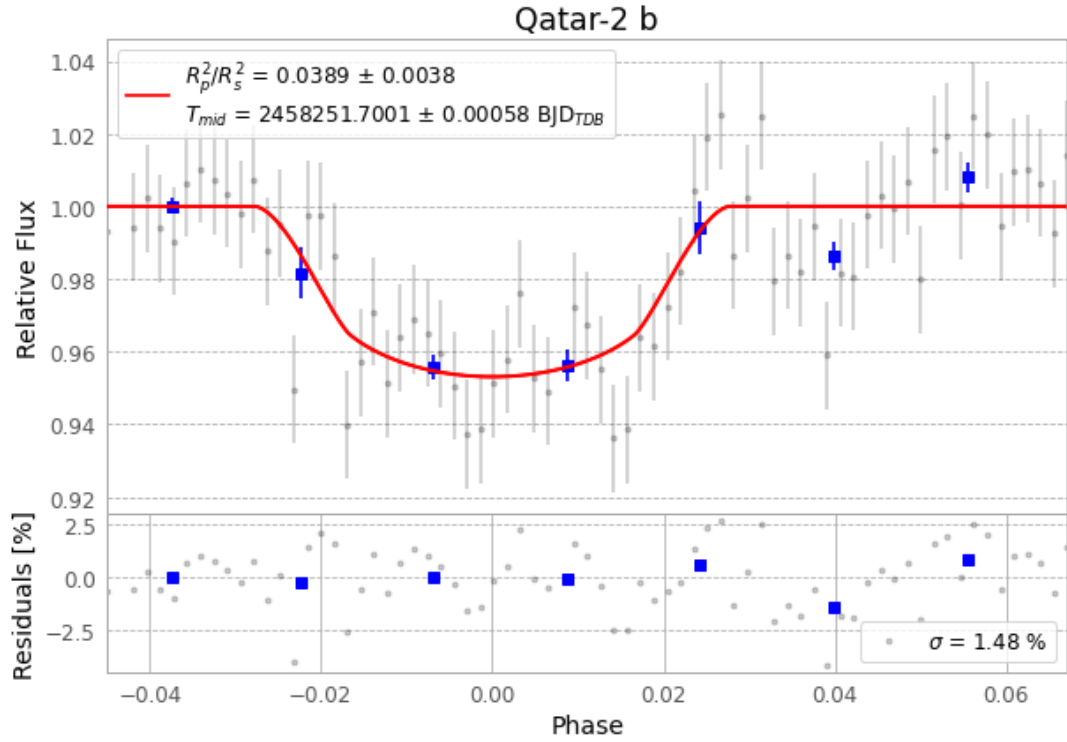


Figure 1 — FinalLightCurve_Qatar-2 b_2018-05-12.png (SHA-256 9fa2dec13c39273e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [489, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a3f9166e2a3909f5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), Qatar-2180513031211.FITS → Qatar-2180513064812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-180513.log (e25a1a44adfd...), FinalLightCurve_Qatar-2 b_2018-05-12.png (9fa2dec13c39...), FinalLightCurve_Qatar-2 b_2018-05-12.csv (28a2431cc5f0...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 11:01Z.